



Lab 1: OLAP

1. Overview

Relational ERDs are sometimes extensively optimized for OLTP (Online Transactional Processing) to avoid problems such as data redundancy and slow writes. On the other hand, OLAP (Online Analytical Processing) systems are optimized for analytical operations that focus on aggregations and groupings on some dimensions or attributes without focusing much on row-level operations.

Therefore, it is always required, for the purpose of OLAP, to convert relational schemas to star schemas. Star schemas are much more optimized for this purpose compared to 3NF schemas, for instance.

Therefore, in this assignment, you will be performing an ETL process to convert some normalized relational schema to a star schema.

2. Installation

As discussed above, it is required to perform an ETL (Extract, Transform, and Load) process to extract relational data from a MySQL data source, transform relational schema into a star counterpart, and finally load the new star schema and query it using Spark.

A) Install MySQL

If you do not have MySQL, refer to this installation guide:

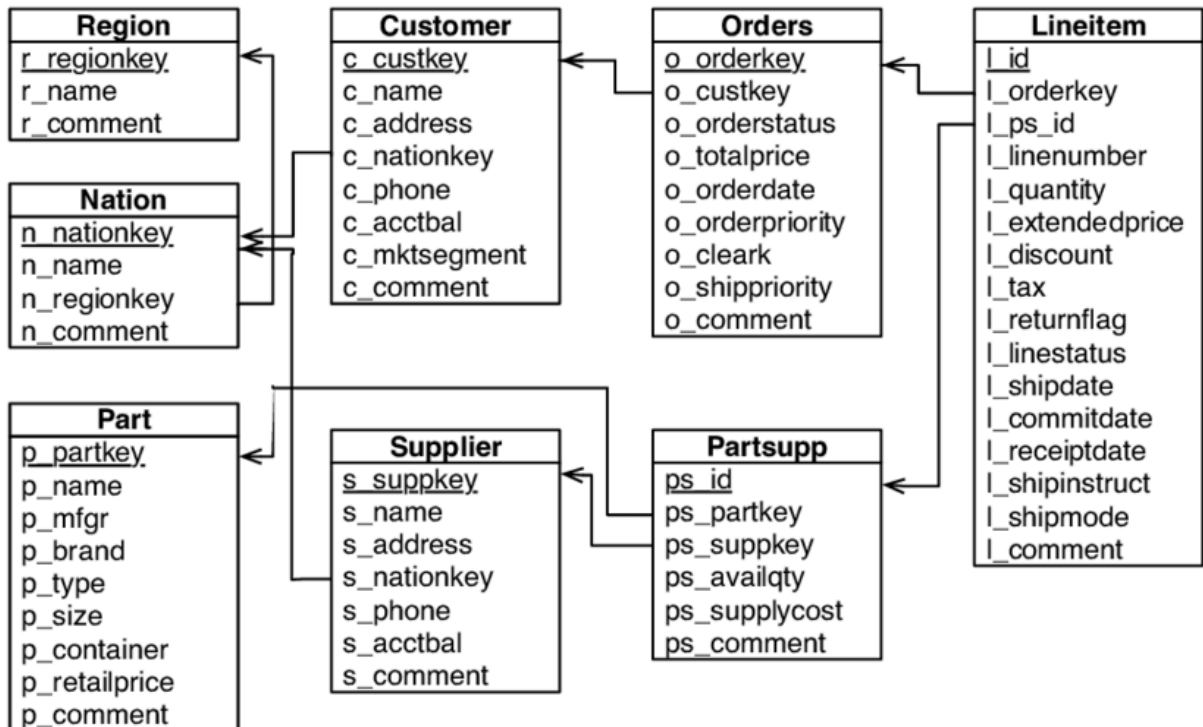
- [Ubuntu 22.04](#)
- [Ubuntu 18.04](#)
- [MacOS](#)



B) Prepare Source OLTP Database

You will start this exercise by importing a ready to use database into MySQL. In this lab, we will use the [TPCH database](#).

Generate TPC data with scale-factor = 1, please check [this tutorial](#) for how to do that



C) Install Apache NiFi

For the ETL part, we will be using Apache NiFi. Check the following to install Apache NiFi for Linux, Mac Os, or Windows. Find these guides for getting started with NiFi on both [linux/Mac OS](#) and [Windows](#). And this is [what you do](#) after starting.

Note: If you install the latest version 2.2.0, and you should, please note that you should use Java 21. Check [this tutorial](#) on how to do this and switch between your current version or 21.



D) Install Apache Spark

For querying the star schema, we will use Spark as an OLAP engine. To install Spark, please follow the instructions in the following guide:

<https://spark.apache.org/downloads.html>

3. Requirements

A) Design Star Schema

Given the following query

```
select
    n_name,
    s_name,
    sum(l_quantity) as sum_qty,
    sum(l_extendedprice) as sum_base_price,
    sum(l_extendedprice * (1 - l_discount)) as sum_disc_price,
    sum(l_extendedprice * (1 - l_discount) * (1 + l_tax)) as sum_charge,
    avg(l_quantity) as avg_qty,
    avg(l_extendedprice) as avg_price,
    avg(l_discount) as avg_disc,
    count(*) as count_order
from
    lineitem,
    orders,
    customers,
    nation,
    partsupp,
    supplier
where
    l_shipdate <= date '1998-12-01' - interval '90' day AND
    l_orderkey = o_orderkey AND
    o_custkey = c_custkey AND
    c_nationkey = n_nationkey AND
    l_ps_id = ps_id AND
    ps_suppkey = s_suppkey
group by
    n_name,
    s_name
```

It is required to optimize our database to execute it efficiently. This query is joining 6 tables. The estimated number of rows of these tables is as follows

- nation 5
- supplier 10,000



- customer 150,000
- partsupp 300,000
- orders 1,500,000
- lineitem 6,000,000

You are required to design a star schema that adapts the TPCCH schema for OLAP purposes. Check [this tutorial](#) for how to design such a schema. This set of slides discusses how to design a star schema for one of MySQL's complex schemas, the Sakila schema. Apply the practices used there to the TPCCH schema.

B) Use NiFi for ETL

You are required to use Apache NiFi for migrating data from the source MySQL schema to Parquet files; each table is represented by a Parquet file(s).

ETL stands for Extract, Transform, and Load. Extraction is the process of pulling data out of some data sources for the purpose of transformation or migration. Transformation is the phase of changing the data in transit. Transformations include filtration, mapping, reducing, etc. Loading refers to persisting data into some final destination with capabilities not available in the original source.

NiFi is a drag-and-drop tool to build systems that automate the flow of data between software systems. It uses the concept of computational graphs, where the processing is done by building a graph of processing units called Processors. Some of the Processors are source Processors without any input needed. Others are just terminal Processors without output edges going out.

In our case, you need the following:

1. **A source Processor to fetch data from the MySQL employees schema**
2. **A group of Processors to transform the source relational schema to a star schema**
3. **A Processor to write output records to Parquet files**

The following materials should be useful for you:

- [this simple tutorial](#) shows how to read data from the MySQL source schema.
- [this simple tutorial](#) shows how to write data as Parquet

NiFi Processors of possible interest to us:

- ExecuteSQL
- GenerateTableFetch (**Note: Depending on the size of your machine, you may need this processor to extract MySQL records in batches**)
- ConvertAvroToParquet
- PutFile
- PutParquet



You can still build the application using other Processors.

Please note that you need to configure batch processing, fetching rows chunk by chunk (for example 1000 rows at a time), to avoid out of memory errors due to large data sets being loaded once in NiFi.

Note: You are required to build the transformation from source schema to star schema inside NiFi. You should not use Apache Spark for such processing. Meaning, you should not just transfer the data from MySQL to Parquet in original schema and do the joins using Spark upon query.

C) Execute the query for the Star and the Relational Schema

You are required to run the query above once on MySQL and again using SparkSQL against the parquet files. Take the average of the running this query 8 times

4. Deliverables

You are required to deliver the following:

1. Steps showing generation of TPC data
2. Steps showing insertion of generated data into MySQL
3. DDL of the OTLP schema in MySQL
4. DDL of the Star schema or equivalent of transformation queries in NiFi or parquet schemas.
5. NiFi ETL graph with a brief description of each processor in the processing graph.
6. Table of the 8 runs on MySQL and Spark showing time taken in each of the 16 runs
7. Report containing the previous deliverables

5. Notes

- You should work in groups of **four**
- You should come prepared for running the whole ETL pipeline during discussion if instructed to. **So please get your environment ready for that.**
- **All team members should be ready to answer any question during discussion**
- **Any cheating will be severely penalized.**